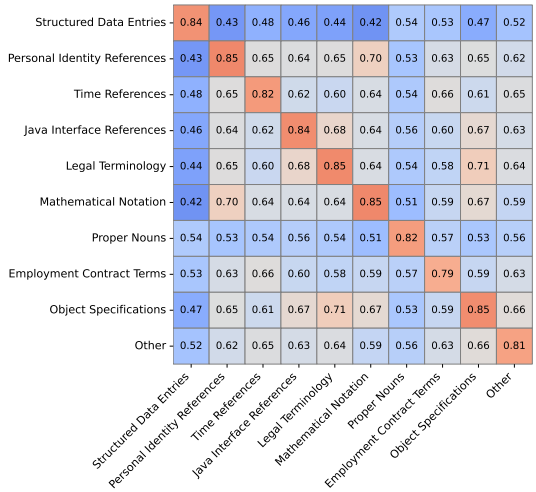
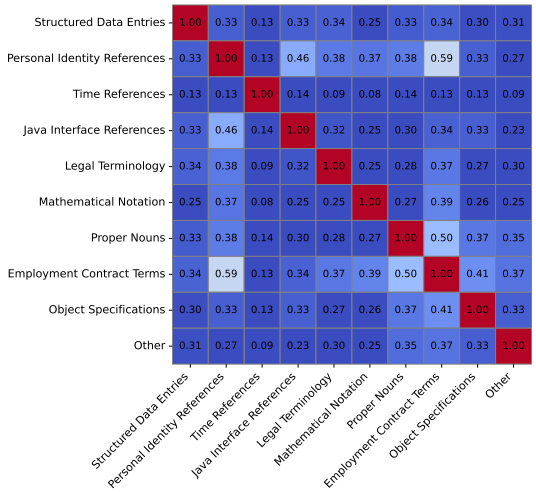


(a) HYPERSTEER (cross attention)



(b) HYPERSTEER (no context)

Figure 6: Pairwise cosine similarities of steering vectors, averaged within each steering prompt, for our two HYPERSTEER variants. (a) **Cross-attention**: strong on-diagonal (same steering prompt) alignment with some off-diagonal variance due to prompt conditioning. (b) **No-context**: generally weaker off-diagonal alignment. We hypothesize that cross-attention's higher off-diagonal similarity arises from shared semantic and linguistic structure across prompts even when steering prompt labels differ.

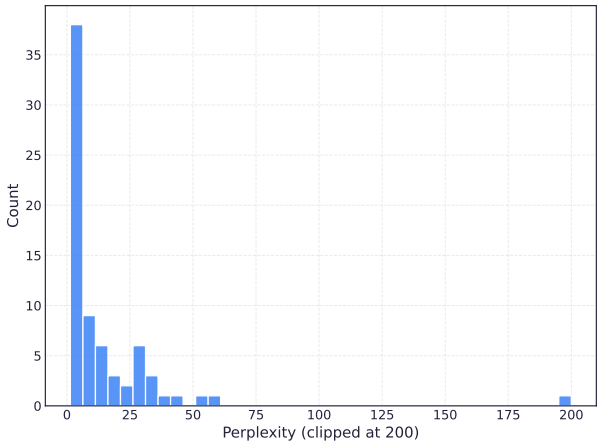


Figure 7: `Concept10` perplexity distribution on data labels sampled from `gpt-4o-mini`.

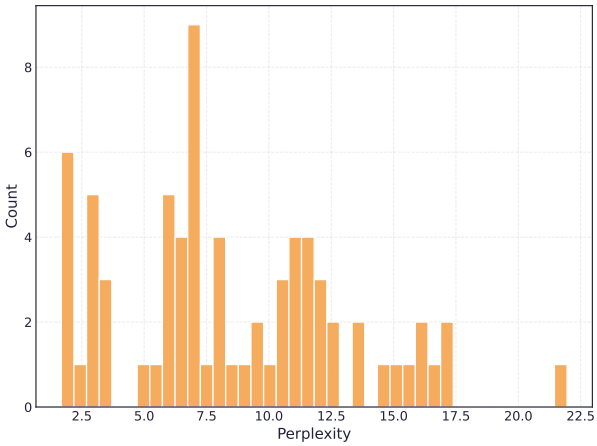


Figure 8: `Concept10` perplexity distribution won data labels sampled from `Gemma-2-2B`.

We ran preliminary experiments by training on labels from both distributions, and find steering performance is still better with the `gpt-4o-mini` labels. We suspect that this could be a result of either lower quality of `Gemma-2-2B` responses due prompt-engineering being the method of generation or the LLM-as-a-judge evaluation setup (which also uses `gpt-4o-mini`) being biased towards outputs from the same model.

A.8 Sample generations

We use the best cross attention model for steering and the steering factor that yields the best aggregate score during evaluation. We also include responses from a prompt steered baseline for comparison. Generation is done with temperature of 1.0 and with multinomial sampling, following AXBENCH. See example generations 13, 14, 15, 16, 17, 18, 19, 20, 21, 22.

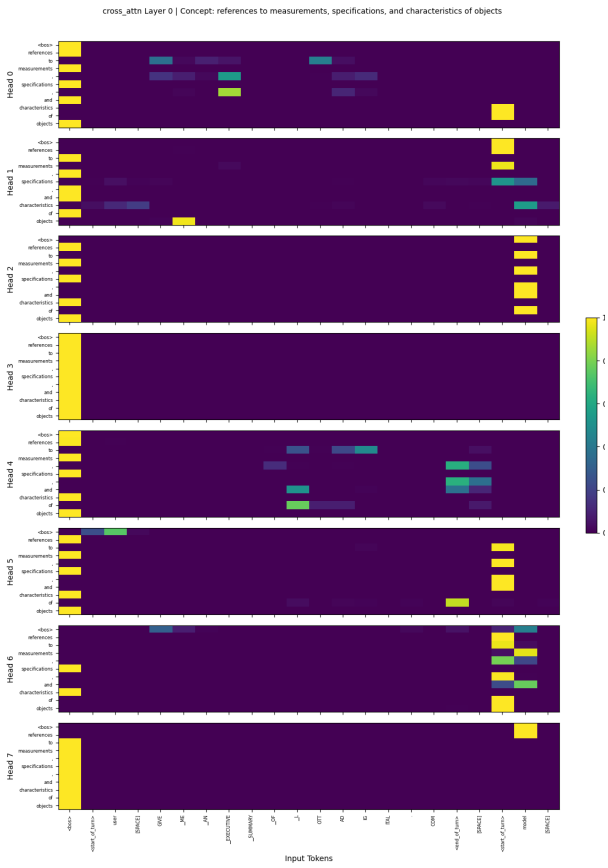


Figure 9: Layer 0 attention map.

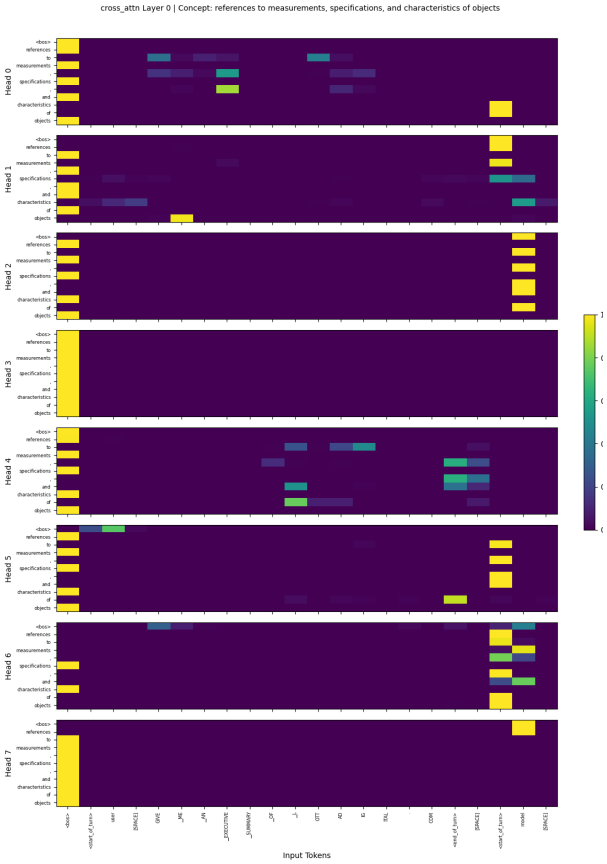


Figure 11: Layer 10 attention map.

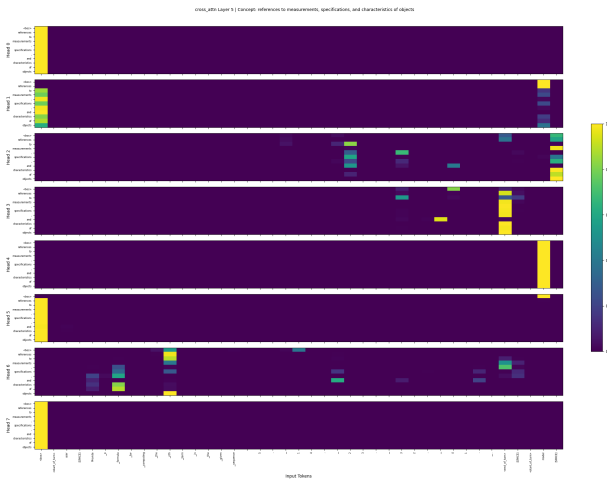


Figure 10: Layer 5 attention map.

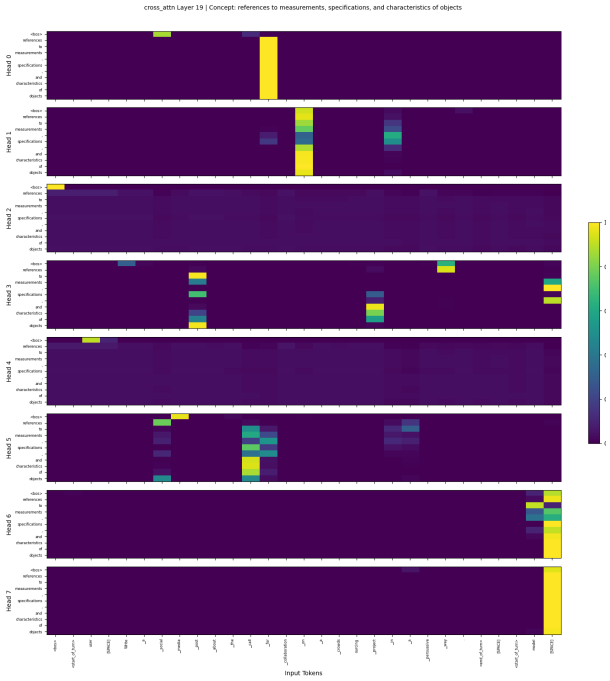


Figure 12: Layer 19 attention map.

Steering Prompt: ***specific identifiers or entries in a structured data format***

Input Prompt

Please give me a recipe I can cook with only the ingredients in my pantry: sausage, peas, yogurt, pasta.

HYPERSTEER Generation

{Recipe Title: Microwave Veggie & Sausage Pasta Ingredients: 1 cup pasta, ½ cup peas (fresh or frozen), 1 link sausage (grated or finely minced), ¼ cup plain yogurt Steps: • Cook pasta per package. • In a microwave-safe bowl, combine peas and sausage; microwave 2 min. • Stir in cooked pasta and yogurt until creamy. • Season with salt/pepper. }

PromptSteering Generation

Spicy Sausage & Pea Pasta with Yogurt Sauce 1. Boil 1 cup pasta until al dente. 2. In a skillet, sauté 1 link sausage (sliced) with ½ cup peas for 3 min. 3. Stir in ¼ cup plain yogurt off heat; season to taste.

Figure 13: Successful steering and instruction following by HYPERSTEER.

Steering Prompt: ***instances of specific time references or moments within the text***

Input Prompt

Please let me know your thoughts on the given description of Barcelona's weather and environment.

HYPERSTEER Generation

Barcelona, in the early summer sun, when the morning breeze drifts off the Mediterranean and the streets glow at dawn, feels both lively and languid. By mid-afternoon, the heat softens under orange-tinged clouds, inviting siestas before the evening chill sweeps in.

PromptSteering Generation

Barcelona's vibrancy and energy are intoxicating at sunrise, winding down by high noon when shops close for siesta, and reborn as twilight settles over La Rambla.

Figure 14: Successful steering and instruction following by HYPERSTEER.

Steering Prompt: ***proper nouns and names***

Input Prompt

Write a LinkedIn post to announce that you've joined [New Company] as a Software Engineer.

HYPERSTEER Generation

Excited to share that I've joined [New Company] as a Software Engineer on the Platform Team! Looking forward to collaborating with the talented folks at [New Company]—big thanks to everyone who supported me along the way.

PromptSteering Generation

Thrilled to announce that I've joined the talented team at [New Company] as a Software Engineer in the Platform division! Grateful for the warm welcome and eager to get started.

Figure 15: Successful steering and instruction following by HYPERSTEER.